

Deep Learning for Explainable Cancer Identification Spectroscopy

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2025

Abstract

This is some stuff

1 Introduction

Cancer is one of the leading causes of death worldwide with around 20 million cases reported in 2022. This corresponds to one in five men or women developing cancer in their lifetime [Bray et al. \[2024\]](#). The conventional method for cancer diagnosis, as outlined in Figure 1, involves taking a biopsy of suspect tissue from a patient during surgery, fixing this biopsy to a slide, and then staining the biopsy with haematoxylin and eosin (H&E). The haematoxylin stains nucleic acids a deep blue-purple colour, whilst the eosin is pink and non-specifically stains proteins [Fischer et al. \[2008\]](#). These H&E stained samples can then be analysed by a pathologist to develop a diagnosis. This process is manual, error-prone, subjective, costly, and time-consuming; often requiring transport to a laboratory for processing by highly trained technicians [Hollon et al. \[2020\]](#). The sample processing time alone is often 20 minutes or more [Novis and Zarbo \[1997\]](#), so the development of methods which forgo traditional lab-based preparation have significant potential to guide surgeries more effectively.

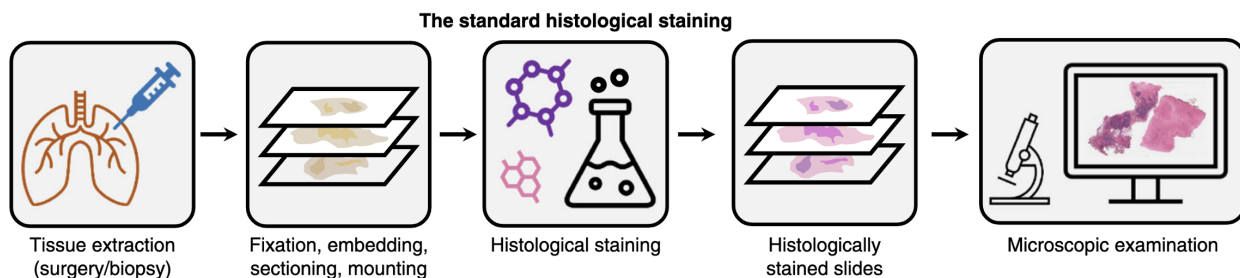


Figure 1: From [Bai et al. \[2023\]](#)

New methods are being developed to expedite the cancer diagnosis process by using machine learning in conjunction with a range of imaging technologies including stimulated Raman spectroscopy (SRS) [Hollon et al. \[2020\]](#), [Sarri et al. \[2019\]](#), [Kondepudi et al. \[2024\]](#), [Jiang et al. \[2022\]](#), second harmonic generation microscopy (SHG) [Sarri et al. \[2019\]](#), and Fourier transform infrared spectroscopy (FTIR) [Tomas et al. \[2022\]](#), [Berisha et al. \[2019\]](#). Within this work, I focus on the application of FTIR spectroscopy to cancer diagnosis within colon cancer samples provided by collaborators at the University of Exeter, which were collected and had FTIR spectral images produced as part of the MINERVA project ?.

1.1 Future Vision

The TROPHY project ? is aiming to develop a new vibrational spectroscopic imaging microscope exploiting photo-thermal infrared (PTIR), FTIR and digital holography microscopy (DMH) to evaluate cell-type topologies, identify mutant cell clusters, and provide biochemical analyses to determine tumour types, grading and staging ?. The MINERVA and TROPHY datasets are expected to be similar, so the models and methods developed in this work constitute a basis for future work towards the TROPHY goals. This is particularly apparent given that TROPHY will focus primarily on the same fingerprint $1000\text{--}1800\text{ cm}^{-1}$ region of the IR spectrum as MINERVA.

Other works which could benefit from the approaches described here include work on the CHARM project which is also taking place in Cambridge. This project is similar to TORPHY in that they are developing a new microscope for cancer diagnosis. CHARM are instead focussing on Raman, SGH and TPEF signals, as these have been shown to have some advantages over FTIR ?. Currently, data produced by that microscope is very noisy, and also only has intensity information at around 20 wavelengths, limiting the viability of the SCNN method implemented in this work. However, as the microscopy performance improves, it should be possible to apply the same methods as described here. Some work was done throughout this project to remove vignetting effects in the CHARM dataset. However, this was only pursued briefly, so information relating to it is attached in appendix ??

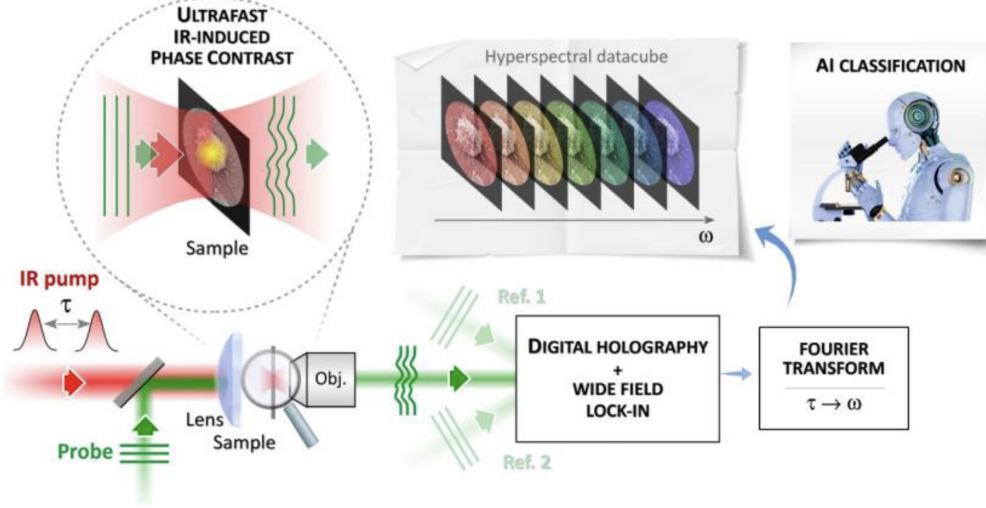


Figure 2: From Trophy proposal [Eic](#)

1.2 Novelty of This Work

This work is focusses on developing a deep learning method for cancer identification from FTIR data which is both explainable and generalisable. This is achieved by first implementing a physically-motivated convolutional neural network, before then utilising a context-representative method to create a latent space from within which different clusters can be identified to elucidate the multiple pathways by which an individual classification such as cancer or non-cancer can be achieved. No other work has been found to use a similar method, although others have used the context-representative method to extract different spectral contributions within urine samples ?. This work then further extends this method, by utilising the CRIME predictions of average spectra spectra and weightings describing the importance of each wavelength within those spectra, to then produce a simple weighted cosine similarity model for classification into non-cancer, hyperplastic, adenoma and cancerous classes.

2 Methods and Results

2.1 FTIR Spectroscopy

Infrared spectroscopy works by directing a broadband spectrum with wavenumbers ranging from around $400 - 4000 \text{ cm}^{-1}$ (for mid-IR spectroscopy) [Su and Lee \[2020\]](#) onto a sample. This coherent spectrum then excites resonant vibrational modes within molecular bonds inside the sample, causing absorption of specific frequencies depending on the composition of the incident sample. This decreases the intensity relative to the baseline in the absorbed frequencies, therefore allowing a characteristic spectrum for each bond to be established. Modern devices use a Fourier transform-based (FTIR) interferogram approach [Griffiths \[1983\]](#), enabling high signal-to-noise (SNR) ratios to be achieved.

FTIR can detect biochemical compositions including nucleic acids, proteins, lipids and carbohydrates within biological samples by matching the observed spectra with known forms for different bonding types,

functional groups and intermolecular interactions [Su and Lee \[2020\]](#). Typical absorption groups within biological samples are detailed in Table 1.

Wavenumber(cm^{-1})	Assignment
3080–2800	Anti-symmetric and symmetric C–H stretches from proteins and lipids
1745–1725	Ester carbonyl of lipids
1700–1500	Amide I and II groups in peptide linkages of proteins
1270–1080	Anti-symmetric and symmetric C–O and P–O areas in DNA, RNA and phospholipids
1200–900	Carbohydrate vibrations of glucose, fructose and glycogen

Table 1: Assignments of Various Functional Groups Based on Wavenumber

2.1.1 Raman Spectroscopy

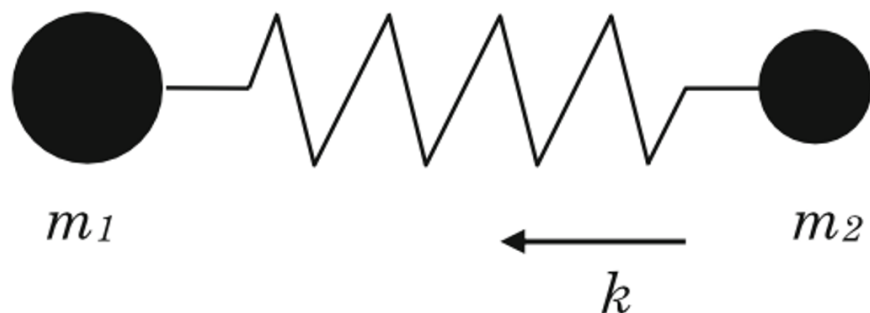
2.1.2 TPEF

2.1.3 SHG

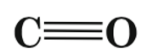
2.2 Classical ML Methods for Spectral Classification

Table 2: Sensitivity and specificity (mean $\pm$ SD) for each model.

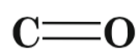
Model	Metric	Normal	Hyperplastic	Adenoma	Cancer	Accuracy
PCA > LDA	Sensitivity	42 $\pm$ 20.4	31 $\pm$ 17.7	84 $\pm$ 14.3	39 $\pm$ 26.3	61 $\pm$ 7.6
	Specificity	90 $\pm$ 11.1	94 $\pm$ 5.4	57 $\pm$ 19.9	95 $\pm$ 6.2	—
XGBoost	Sensitivity	58 $\pm$ 11.9	41 $\pm$ 14.7	80 $\pm$ 14.4	62 $\pm$ 33.6	68 $\pm$ 10.0
	Specificity	88 $\pm$ 6.6	91 $\pm$ 2.6	77 $\pm$ 10.2	96 $\pm$ 6.2	—
PCA > XGBoost	Sensitivity	43 $\pm$ 11.5	28 $\pm$ 11.9	55 $\pm$ 11.9	75 $\pm$ 20.9	51 $\pm$ 9.2
	Specificity	88 $\pm$ 10.3	90 $\pm$ 6.7	79 $\pm$ 7.7	77 $\pm$ 10.5	—



Effect of chemical bonds :



$\sim 2100 \text{ cm}^{-1}$

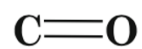


$\sim 1680\text{-}1630 \text{ cm}^{-1}$



$\sim 1200 \text{ cm}^{-1}$

Effect of H-bonding :

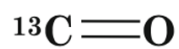


$\sim 1680\text{-}1630 \text{ cm}^{-1}$

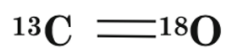


$\Delta\nu \sim - (10\text{-}20) \text{ cm}^{-1}$

Effect of mass :



$\Delta\nu \sim - (35\text{-}55) \text{ cm}^{-1}$



$\Delta\nu \sim - 65 \text{ cm}^{-1}$

Figure 3: From [Berthomieu and Hienerwadel \[2009\]](#)

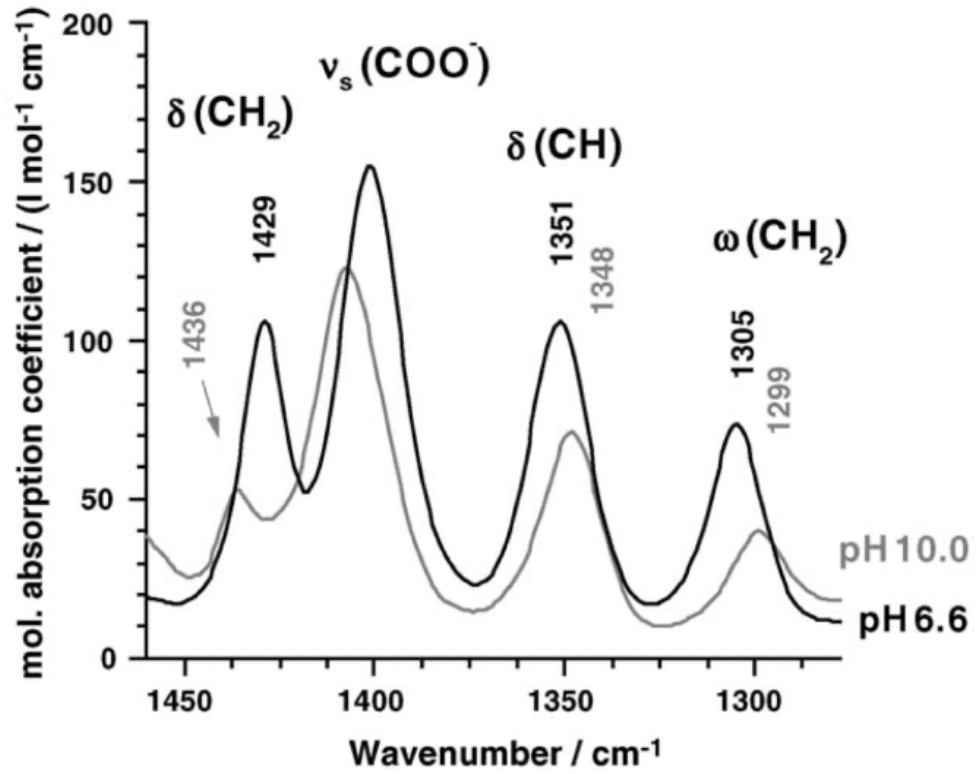
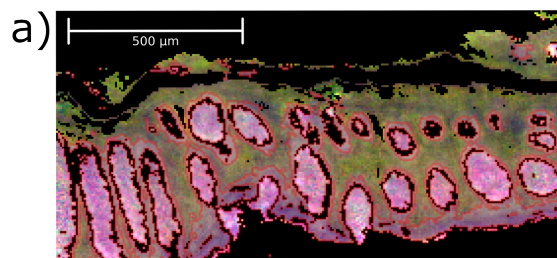


Figure 4: From [Wolpert and Hellwig \[2006\]](#)

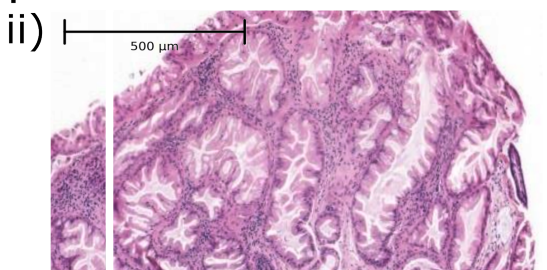
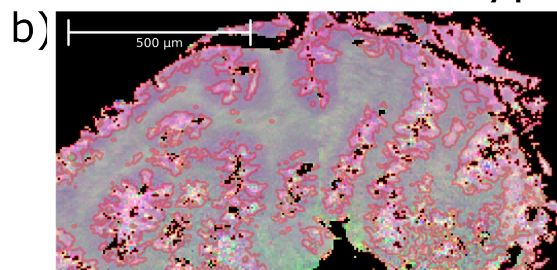
Table 3: Metrics for single fold test set with 50% voting threshold and no label smoothing

Metrics	Normal	Hyperplastic	Adenoma	Cancer
Accuracy	0.699	0.680	0.699	0.945
Sensitivity	0.662	0.258	0.452	0.783
Specificity	0.708	0.762	0.936	0.972
False Positive Rate	0.292	0.238	0.064	0.028
Overall Weighted Accuracy: 0.5387				

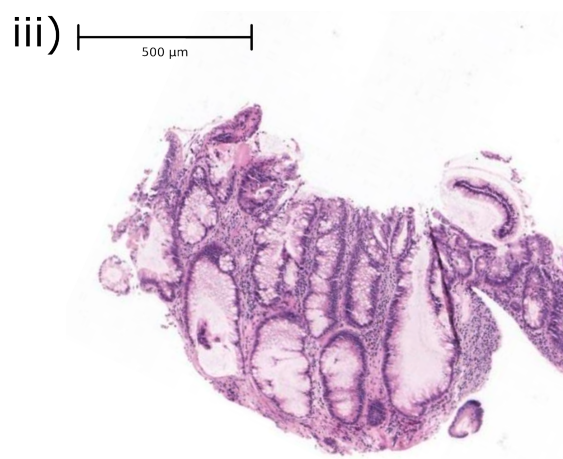
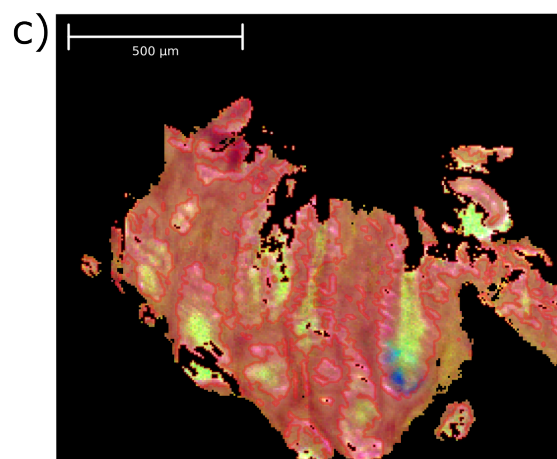
Normal



Hyperplastic



Adenoma



Cancer

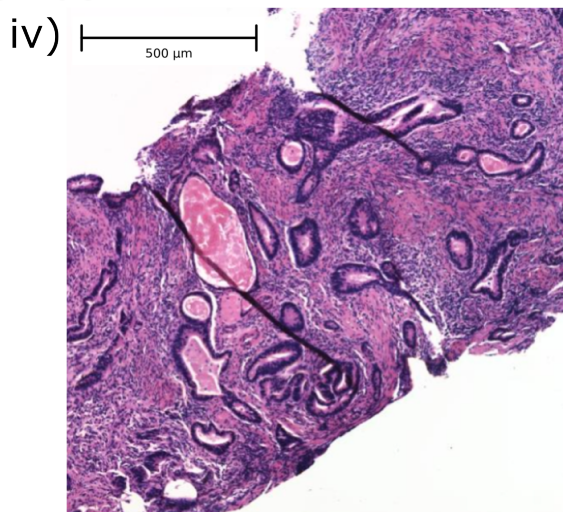
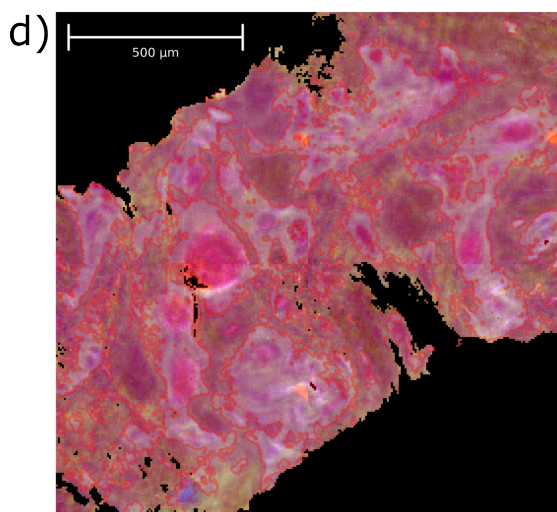


Figure 5: Caption

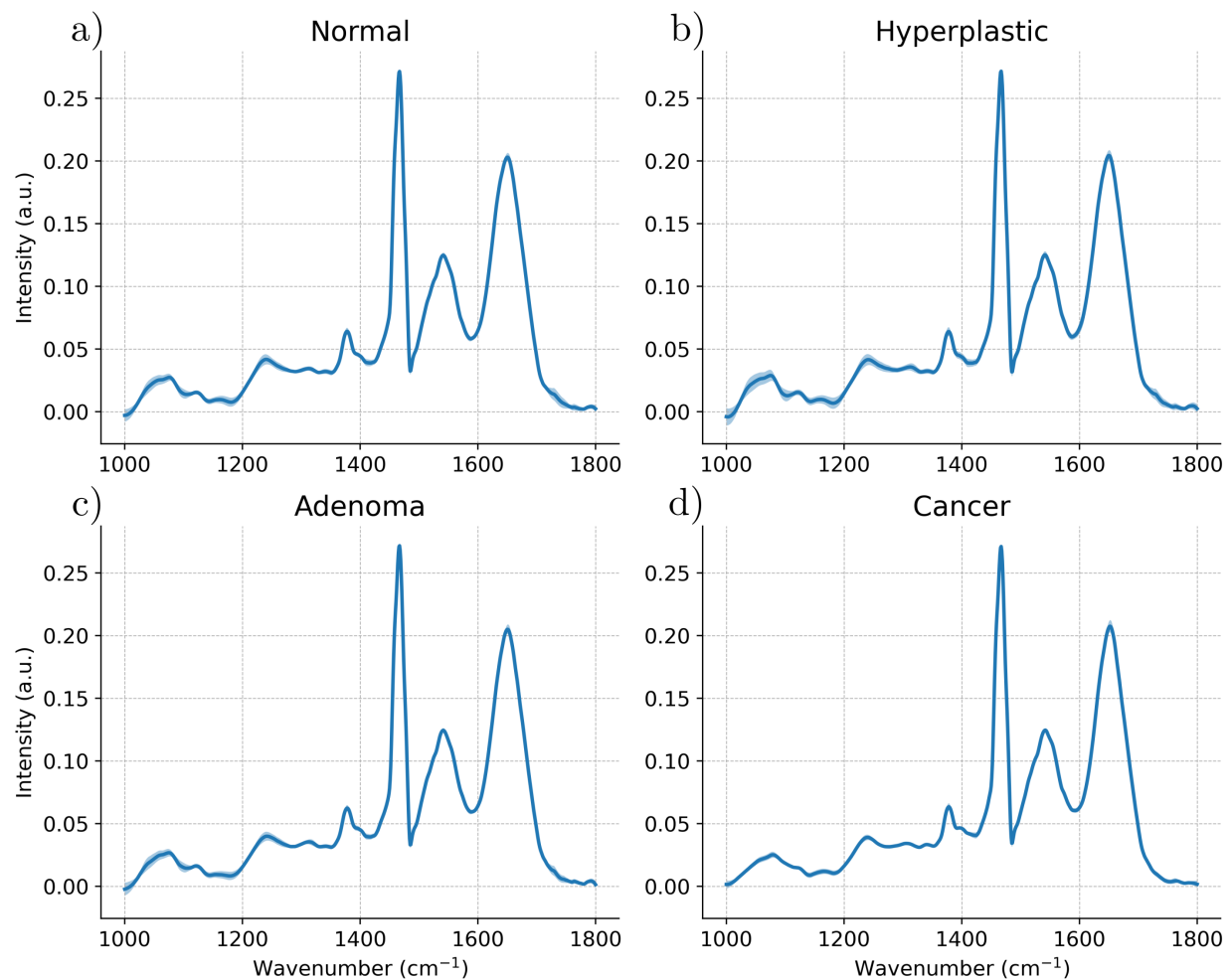


Figure 6: Caption

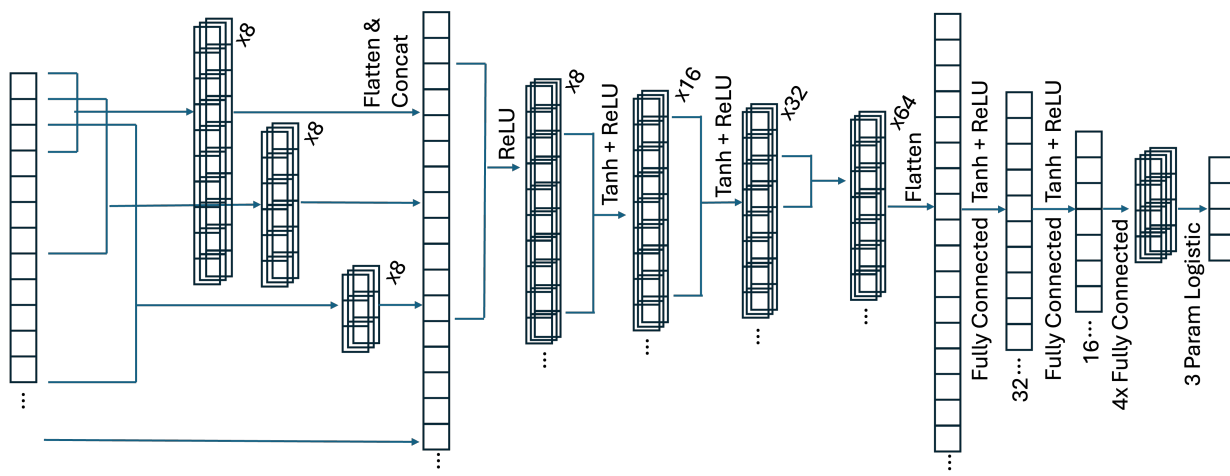


Figure 7: Caption

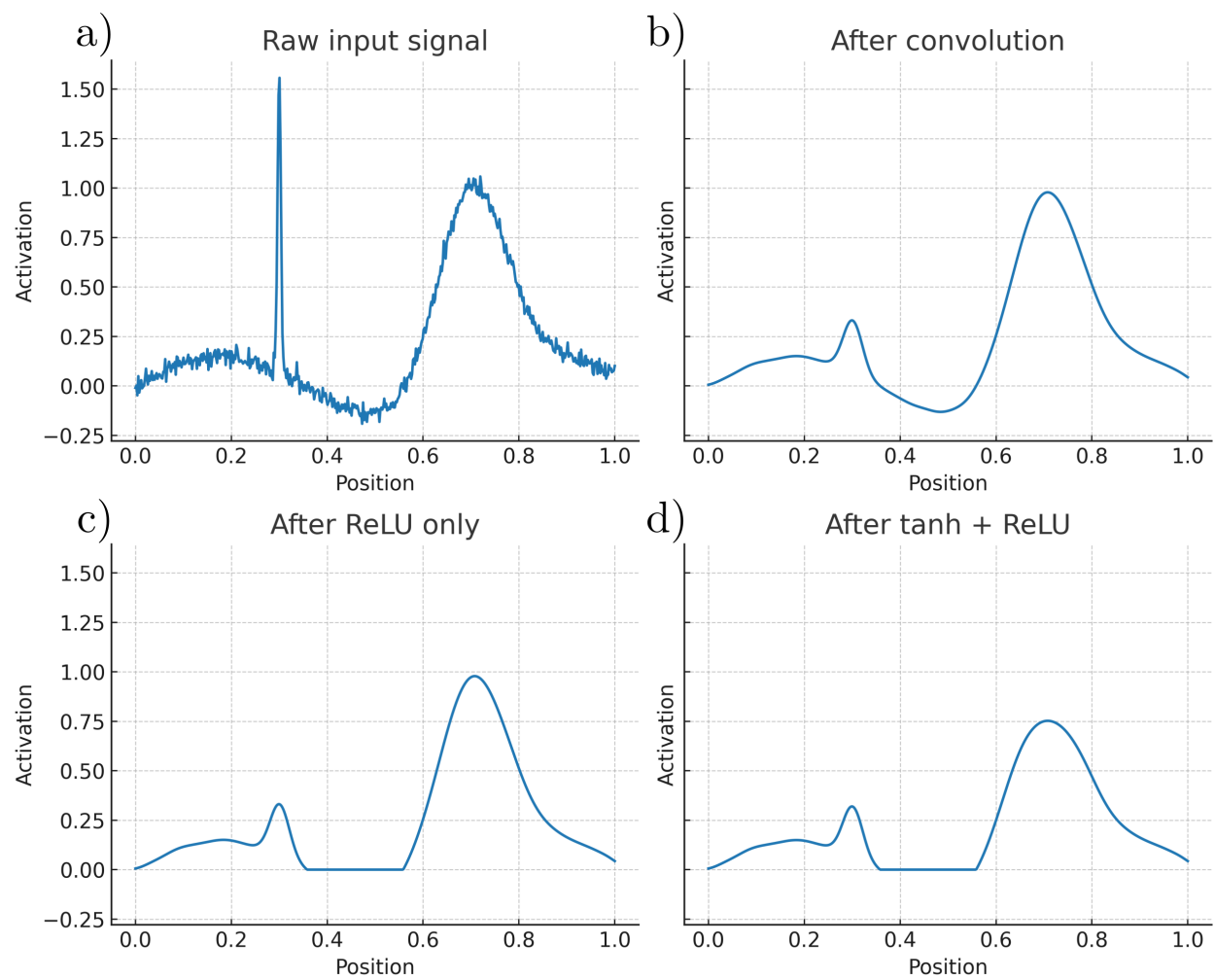


Figure 8: Caption

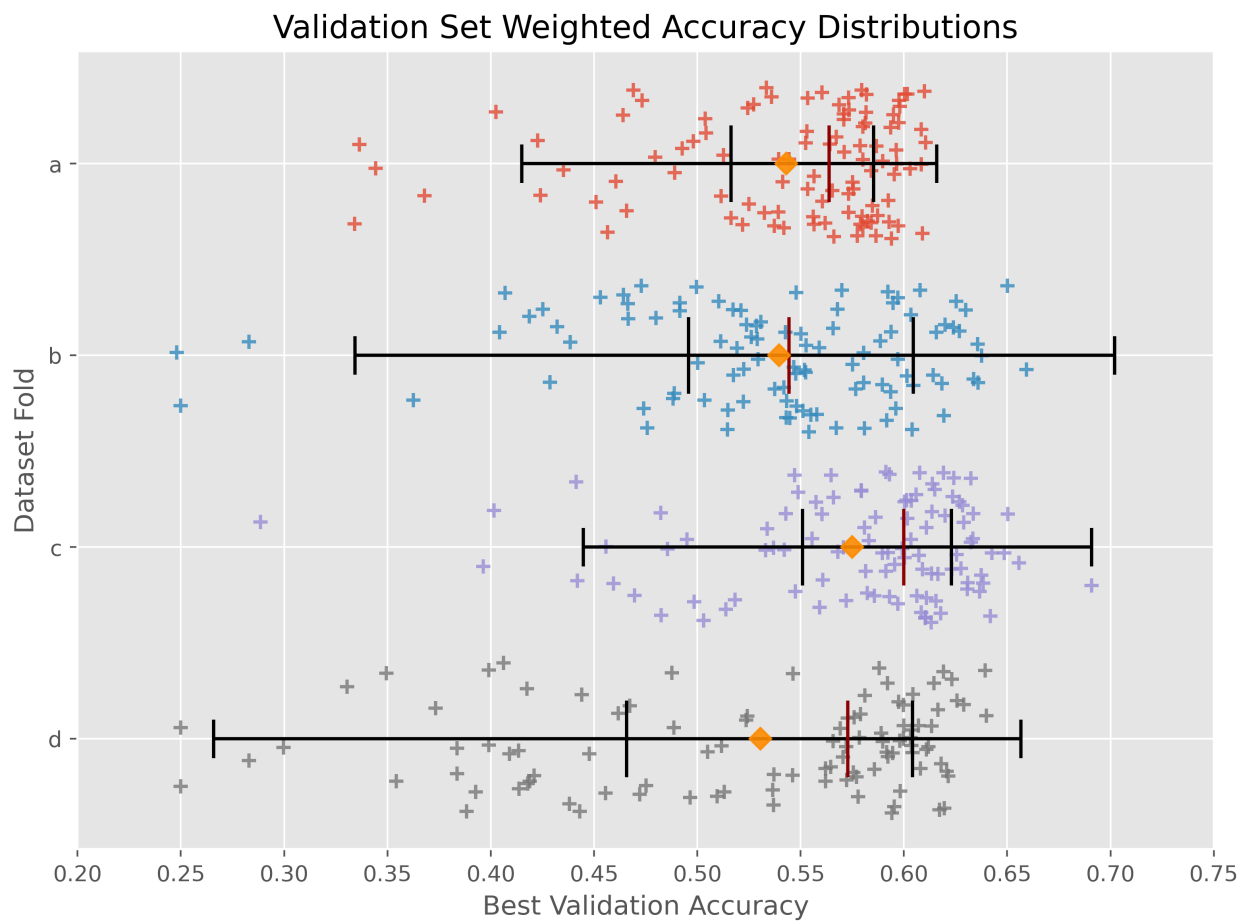


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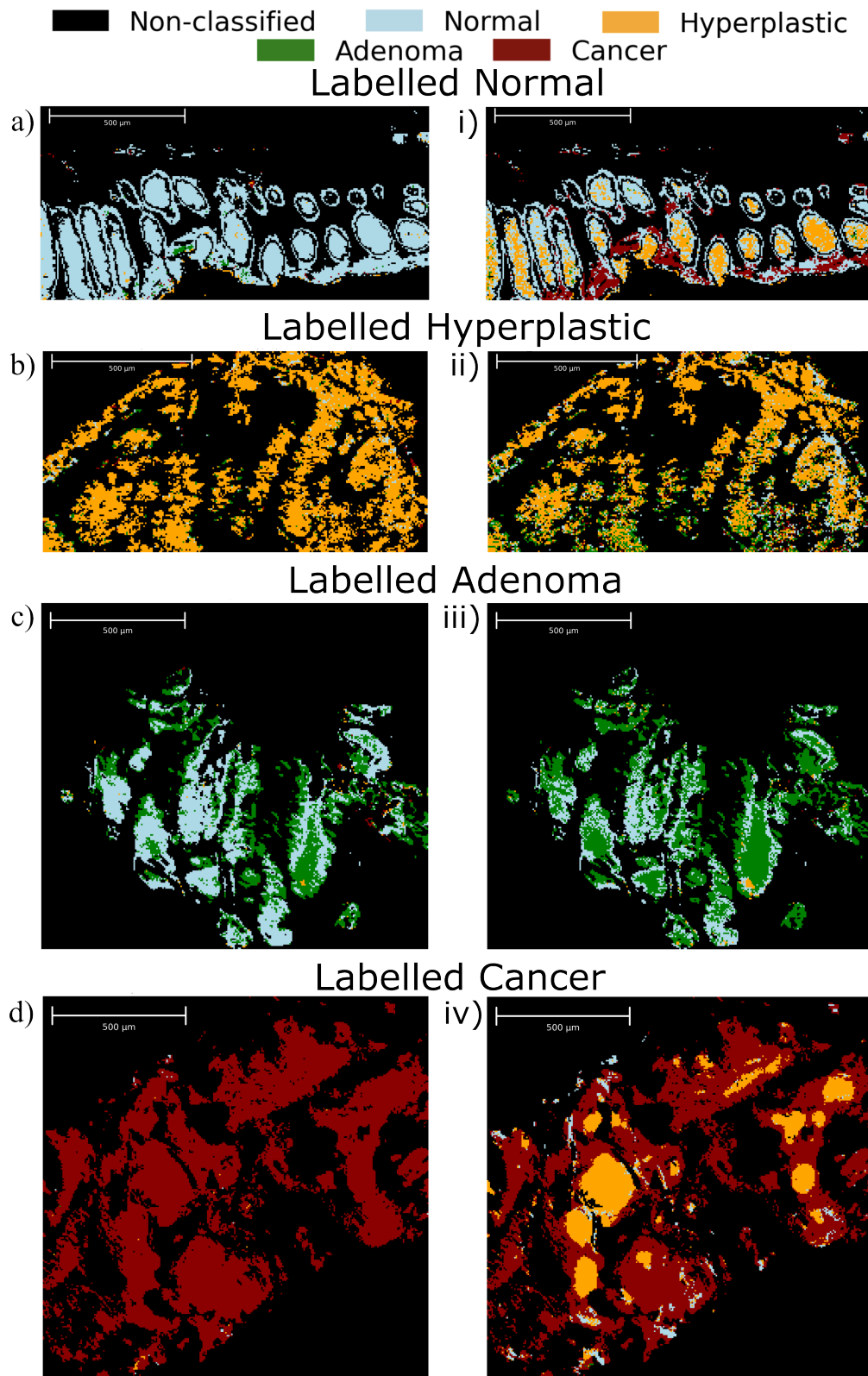


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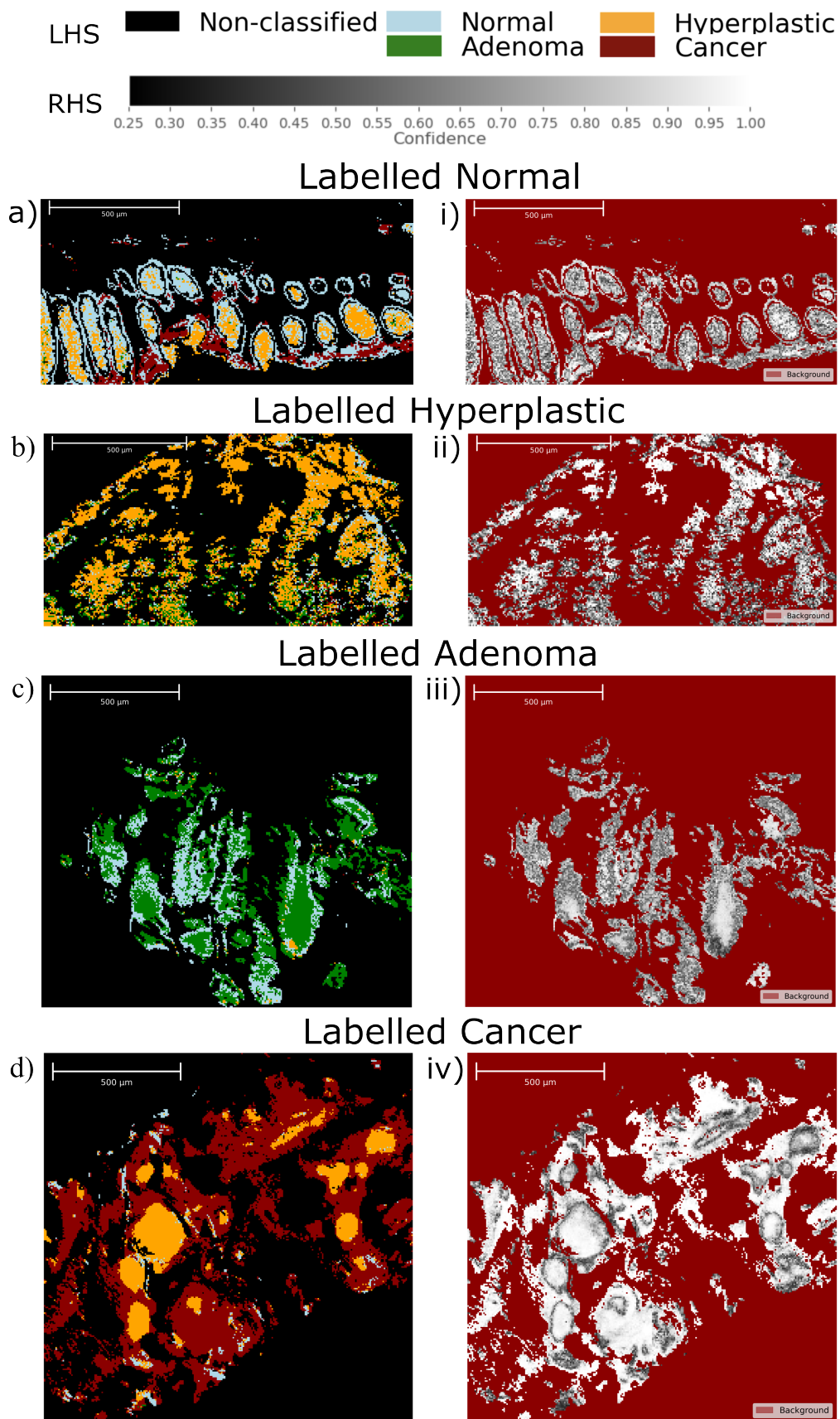


Figure 11: Caption

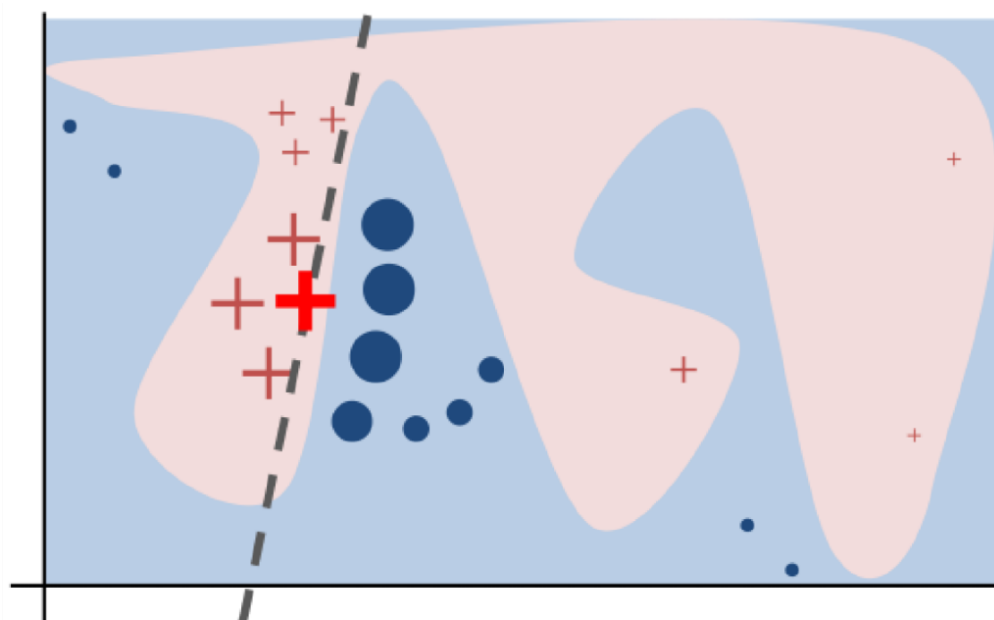


Figure 12: From [Ribeiro et al. \[2016\]](#)

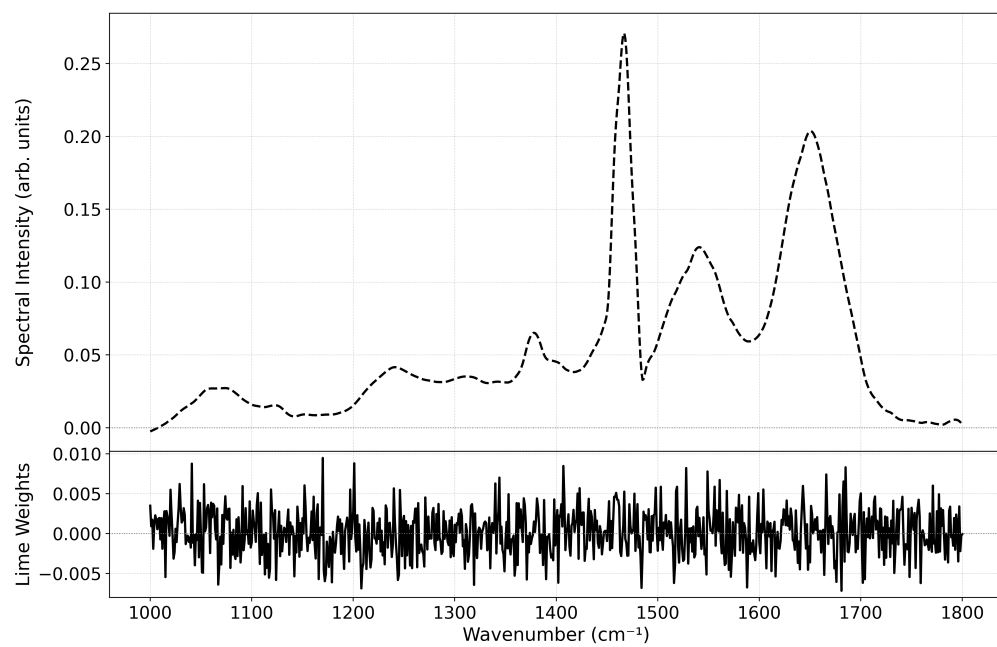


Figure 13: Caption

Table 4: Metrics for single fold test set with 50% voting threshold and 10% label smoothing

Metrics	Normal	Hyperplastic	Adenoma	Cancer
Accuracy	0.702	0.674	0.700	0.944
Sensitivity	0.643	0.257	0.458	0.782
Specificity	0.716	0.755	0.933	0.971
False Positive Rate	0.284	0.245	0.067	0.029
Overall Weighted Accuracy: 0.5349				

Table 5: Metrics for single fold test set with 50% voting threshold and 20% label smoothing

Metrics	Normal	Hyperplastic	Adenoma	Cancer
Accuracy	0.700	0.672	0.692	0.941
Sensitivity	0.650	0.266	0.436	0.786
Specificity	0.713	0.751	0.937	0.967
False Positive Rate	0.287	0.249	0.063	0.033
Overall Weighted Accuracy: 0.5345				

2.2.1 XGBoost

2.2.2 Spectral Matching

2.3 Deep Learning Methods for Spectral Classification

2.3.1 Dataset Challenges

2.3.2 De-noising Methods

2.3.3 Evaluation Metrics

2.3.4 Model Architecture

2.3.5 Regularisation Techniques

2.3.6 Model Performance

2.4 Explainability

2.4.1 Local Interpretable Model-Agnostic Explanations

2.4.2 CRIME

2.4.3 VAE Methods

Clustering VAE Approaches

Logic Explained Networks

3 Results

4 Discussion

5 Conclusions

References

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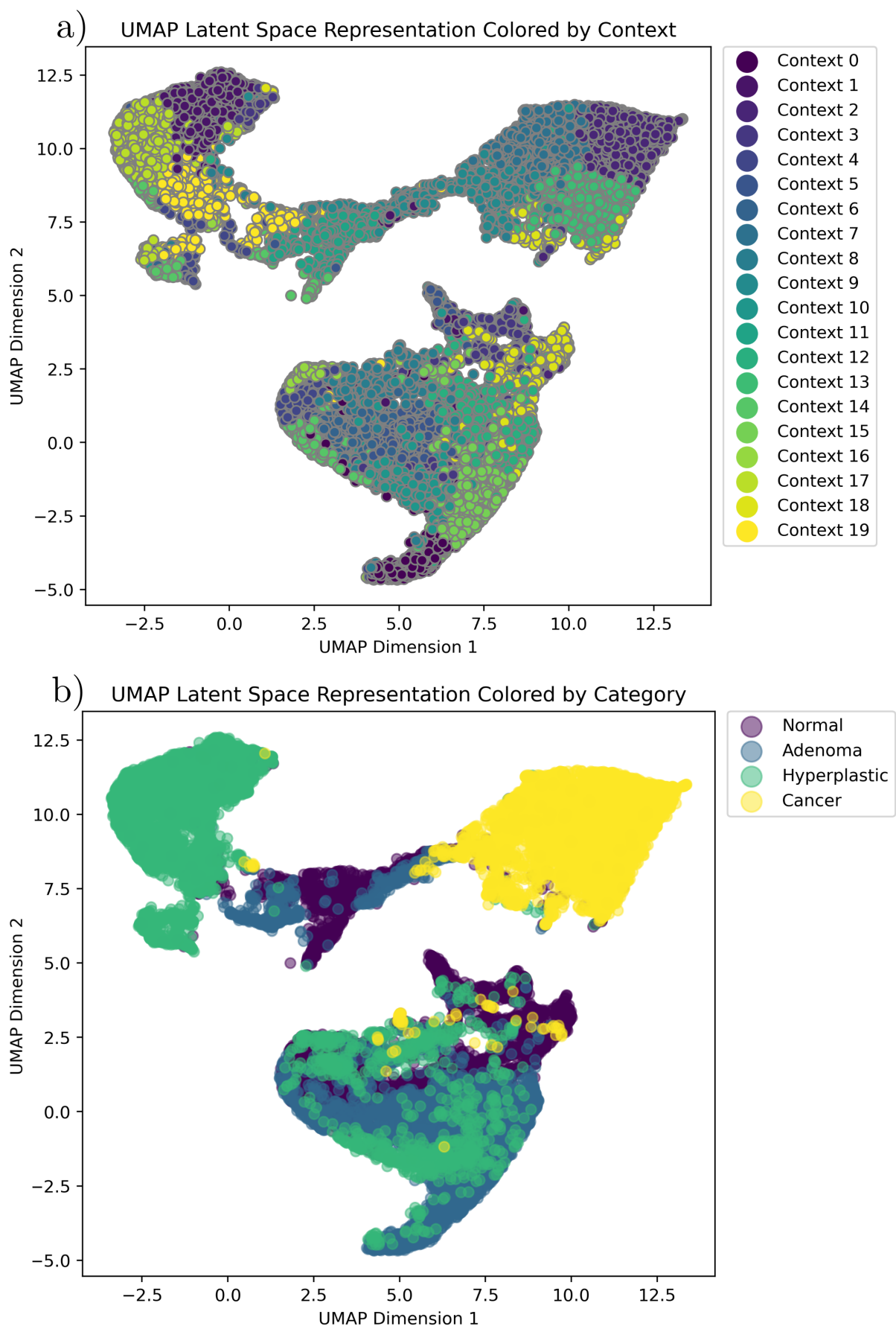


Figure 14: Caption

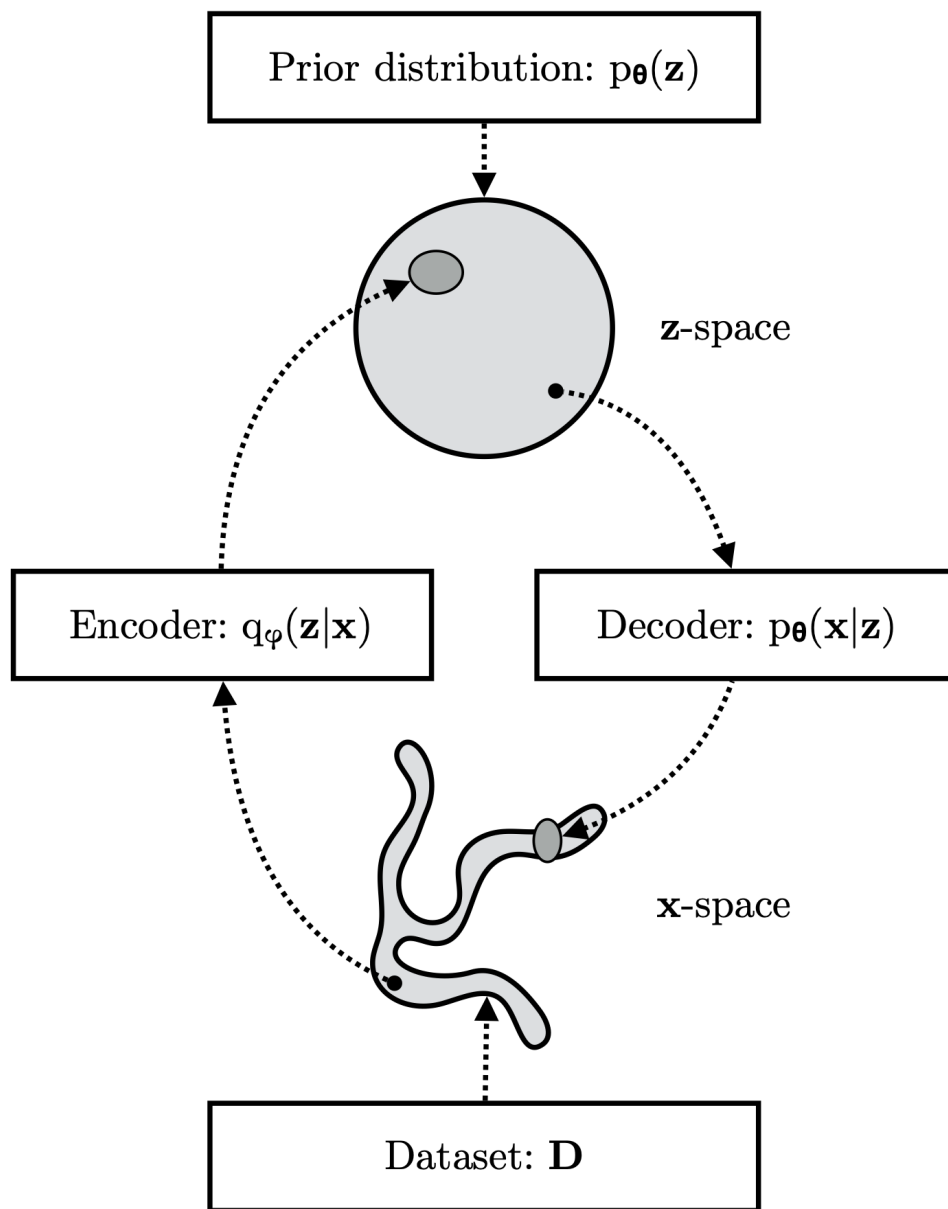


Figure 15: From [Kingma and Welling](#)

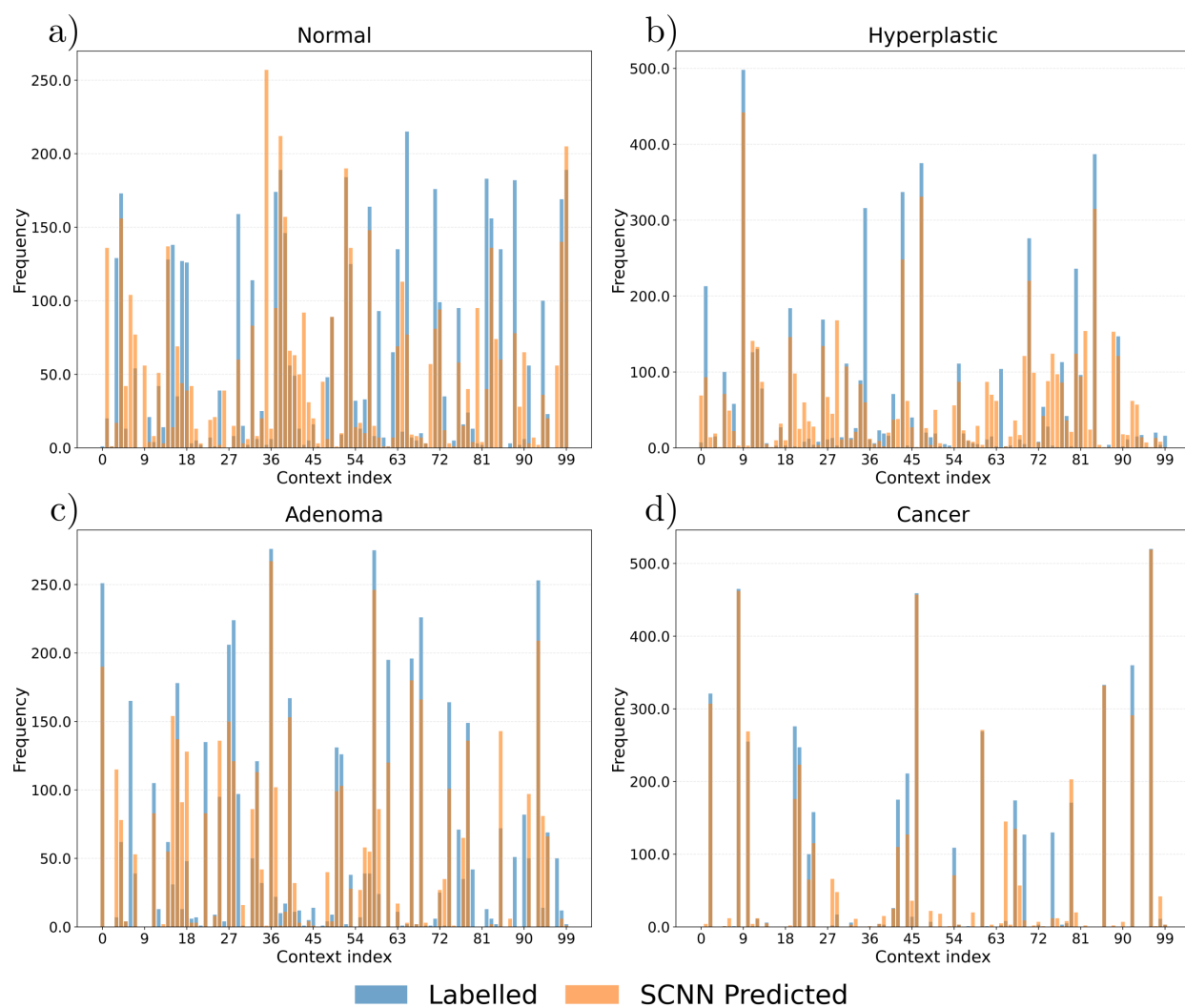


Figure 16: Caption

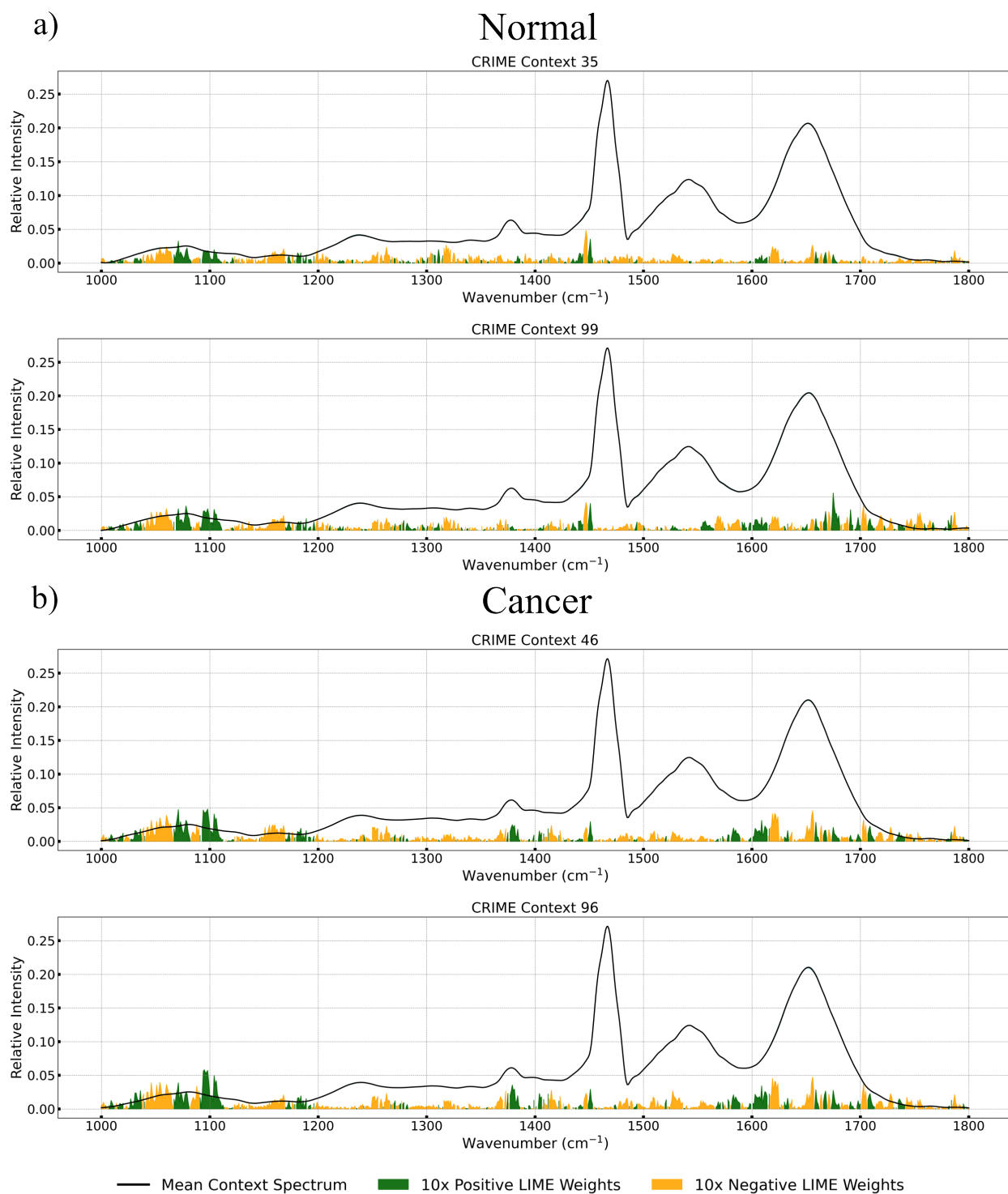


Figure 17: Caption

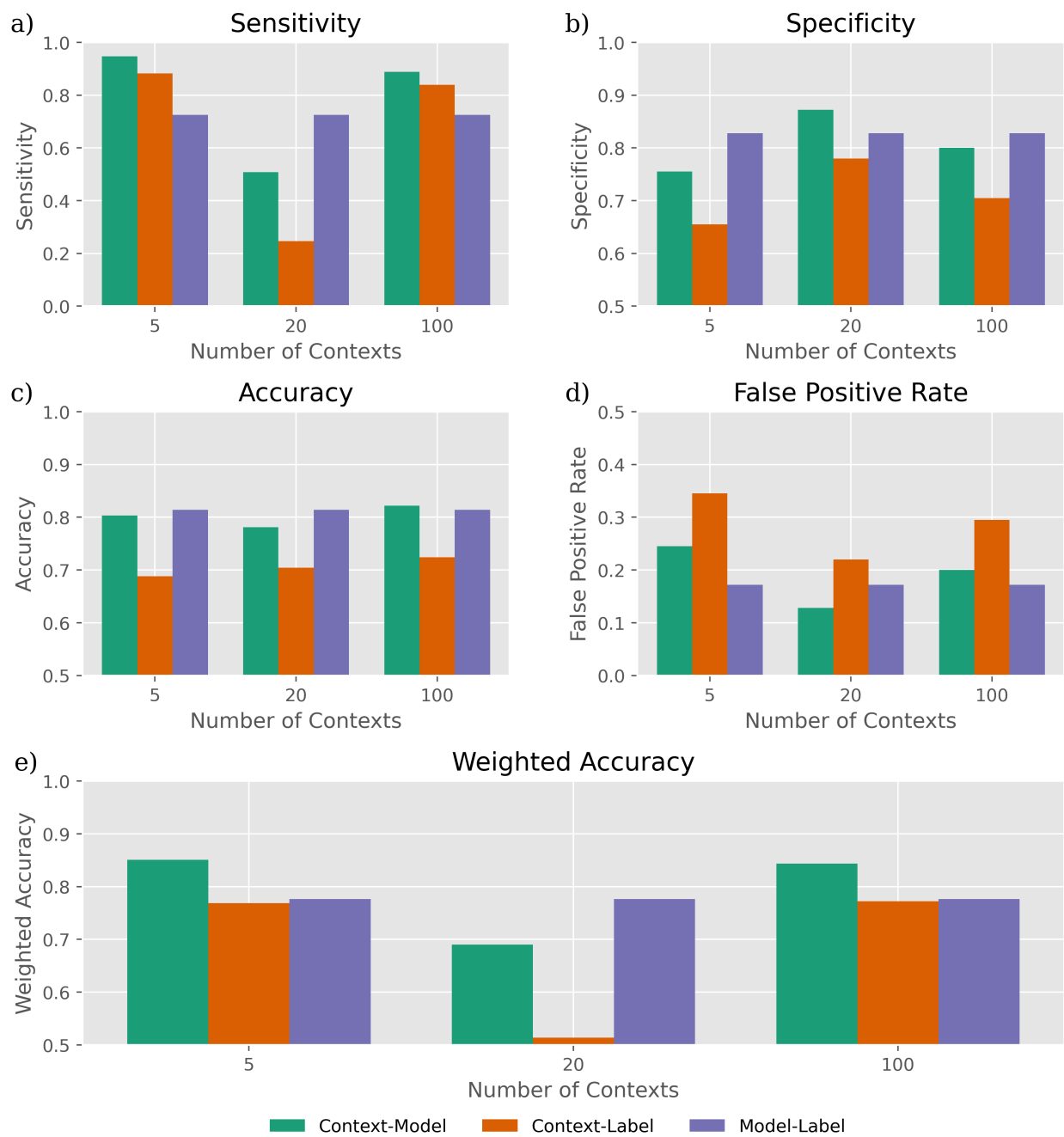


Figure 18: Caption

Table 6: XGBoost

Metrics	Normal	Hyperplastic	Adenoma	Cancer
Accuracy	0.809 ± 0.026	0.826 ± 0.011	0.749 ± 0.073	0.888 ± 0.055
Sensitivity	0.490 ± 0.115	0.386 ± 0.133	0.781 ± 0.055	0.603 ± 0.091
Specificity	0.886 ± 0.029	0.911 ± 0.036	0.711 ± 0.104	0.937 ± 0.057
False Positive Rate	0.115 ± 0.029	0.090 ± 0.036	0.289 ± 0.104	0.063 ± 0.057
Overall Weighted Accuracy: 0.565 ± 0.058				

Table 7: SCNN

Metrics	Normal	Hyperplastic	Adenoma	Cancer
Accuracy	0.756 ± 0.071	0.801 ± 0.083	0.707 ± 0.017	0.881 ± 0.070
Sensitivity	0.739 ± 0.111	0.323 ± 0.069	0.527 ± 0.071	0.777 ± 0.082
Specificity	0.760 ± 0.101	0.894 ± 0.095	0.883 ± 0.089	0.899 ± 0.072
False Positive Rate	0.240 ± 0.102	0.107 ± 0.095	0.118 ± 0.089	0.102 ± 0.072
Overall Weighted Accuracy: 0.591 ± 0.003				

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